Capstone Project Concept Note and Implementation Plan

Project Title: Adaptive Digital Wellbeing Coach: Personalized Prediction of Elevated Smartphone Use on Android

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Concept Note

1. Project Overview

This project develops an Android application that learns each adult user's own smartphone-use patterns, anticipates periods of elevated use and offers a brief, dismissible reminder that encourages reflection. It contributes to Sustainable Development Goal 3 (Good Health and Well-being). Target 3.4 aims, by 2030, to reduce premature mortality from non-communicable diseases by one third and to promote mental health and well-being (United Nations, n.d.); our contribution lies only in the second part, as preventive awareness and self-regulation support. The application does not diagnose or treat smartphone addiction and makes no claim to reduce mortality.

The problem we address is that common tools rely on fixed limits and timers that treat all users alike (Google, n.d.-a), while smartphone behaviour differs greatly between people. In Türkiye, 92.3% of people aged 16–74 used the internet in 2026 (TÜİK, 2026); this measures access rather than problematic use, but it shows how widespread everyday digital habits are. In an exploratory check on two participants of the public dataset described below, daily screen-active time was similar (about 103 versus 92 minutes), yet unlocks per day differed more than threefold (61.5 versus 18.9). A single fixed rule of six unlocks per 30-minute window therefore flagged 5.9% of one participant's windows and 0.4% of the other's. A new user also has no history, so a personalized system faces a "cold-start" problem.

The expected impact is earlier, more personally relevant reminders with fewer false alerts. This is a hypothesis to be tested against simple baselines and, in a small pilot, against users' own evaluations; it is not assumed.

2. Objectives

1. **Reproducible pipeline.** Convert raw event logs into sessions, 30-minute windows and a small feature set that can be computed identically on the public dataset and on the Android prototype.
2. **Documented proxy label.** Define elevated-use indicators (long sessions, repeated unlocking, night-time use), report their prevalence and test their sensitivity to threshold choices.
3. **Model comparison.** Compare a general model, a personalized model and a personal-only model with 1, 3, 7 and 14 days of personal history, using precision, recall, F1, precision–recall AUC and false alerts per day.
4. **Android prototype.** Collect usage events with the user's consent, extract features locally, show brief reminders and ask one feedback question about whether recent use lasted longer than intended.
5. **Evaluation plan.** Check in a small adult pilot whether proxy predictions agree with users' own evaluations, and report feasibility findings if feedback proves too sparse.

Together these objectives address the problem by testing whether information learned from many users improves early personalization, rather than by assuming that adaptive reminders reduce excessive use.

3. Background

Higher smartphone-use severity has been associated with depression, anxiety and poorer sleep in Turkish university students (Demirci et al., 2015), although the cross-sectional design does not show cause and effect. Measurement is also difficult: self-reported and logged use agree only moderately (Parry et al., 2021), so both recorded behaviour and the user's own evaluation are needed.

Existing solutions include Google's Digital Wellbeing, which offers usage summaries, app timers, Focus mode and Bedtime mode (Google, n.d.-a). Socialize helps users choose habits to avoid and alternative activities; it was assessed with 20 users (Monge Roffarello & De Russis, 2021). Time2Stop predicts opportune intervention moments, explains its reminders and adapts to feedback; in an eight-week study with 71 users the effects differed across outcome measures, and the authors identify the cold-start problem as a limitation (Orzikulova et al., 2024).

Machine learning is useful here for three reasons. First, behaviour differs between individuals, so a single threshold performs unevenly. Second, anticipating elevated use means combining several weak signals, such as time of day, recent session length and unlock frequency; tree-based models handle such tabular data well and report feature importance. Third, personalization needs a population model to start from and each user's own history to refine it. Machine learning is not assumed to outperform simple rules; a fixed-rule baseline is always reported.

4. Methodology

The approach has four stages, summarized in Table 1: reconstruct behaviour from event logs, define the prediction target, train and compare models, and explain predictions in plain language.

Table 1. Methodology overview

Stage	Technique	Purpose
Session reconstruction	Unlock-to-screen-off sessions from event logs ordered by raw timestamp; recording gaps flagged, not treated as non-use (Parry & Toth, 2025)	Reliable usage measures
Windowing	Fixed 30-minute windows; features of a completed window predict the proxy outcome of the next window	Anticipation without label leakage
Features	Screen-active time, unlock count, app switches (keyboard and launcher events excluded), previous session length, hour, weekday, night activity, deviation from the user's own history	Low-dimensional and reproducible on Android
Baselines	Fixed rule; logistic regression	Show whether machine learning adds value
Main models	Random Forest and XGBoost	Non-linear tabular modelling with feature importance
Personalization	Pooled retraining with higher sample weight on the current user's earlier records; personal baselines as features	Explicit, reproducible adaptation
Explanation	SHAP TreeExplainer for tree models (SHAP developers, n.d.)	Plain-language reasons, not causal claims

Evaluation protocol. The general model is evaluated with cross-validation grouped by participant, so no participant appears in both training and test folds. For personalization, each held-out participant's first N days with recorded data (N = 1, 3, 7, 14) serve as personal history and the following seven days as the test period, with a gap of one window so that feature and outcome intervals never overlap. Participants with fewer than N + 7 recorded days are excluded from that condition and the counts are reported. The personalized model is trained on the other participants' data plus the user's history, with the weight on the user's records tuned on validation participants only; the personal-only model uses the user's history alone. Class weighting or resampling is applied to training data only. We report precision, recall, F1, precision–recall AUC and false alerts per day, per user as well as on average.

Scope of the pilot. The app pilot will last about one week, which is too short to support claims about 14-day personalization. The personalization curves are therefore computed on the public dataset, and the pilot serves to test feasibility, collect feedback labels and compare proxy predictions with users' answers. If time permits, an LSTM sequence model (Katevas et al., 2017) will be tested as an extension; it is not part of the first phase.

5. Architecture Design Diagram

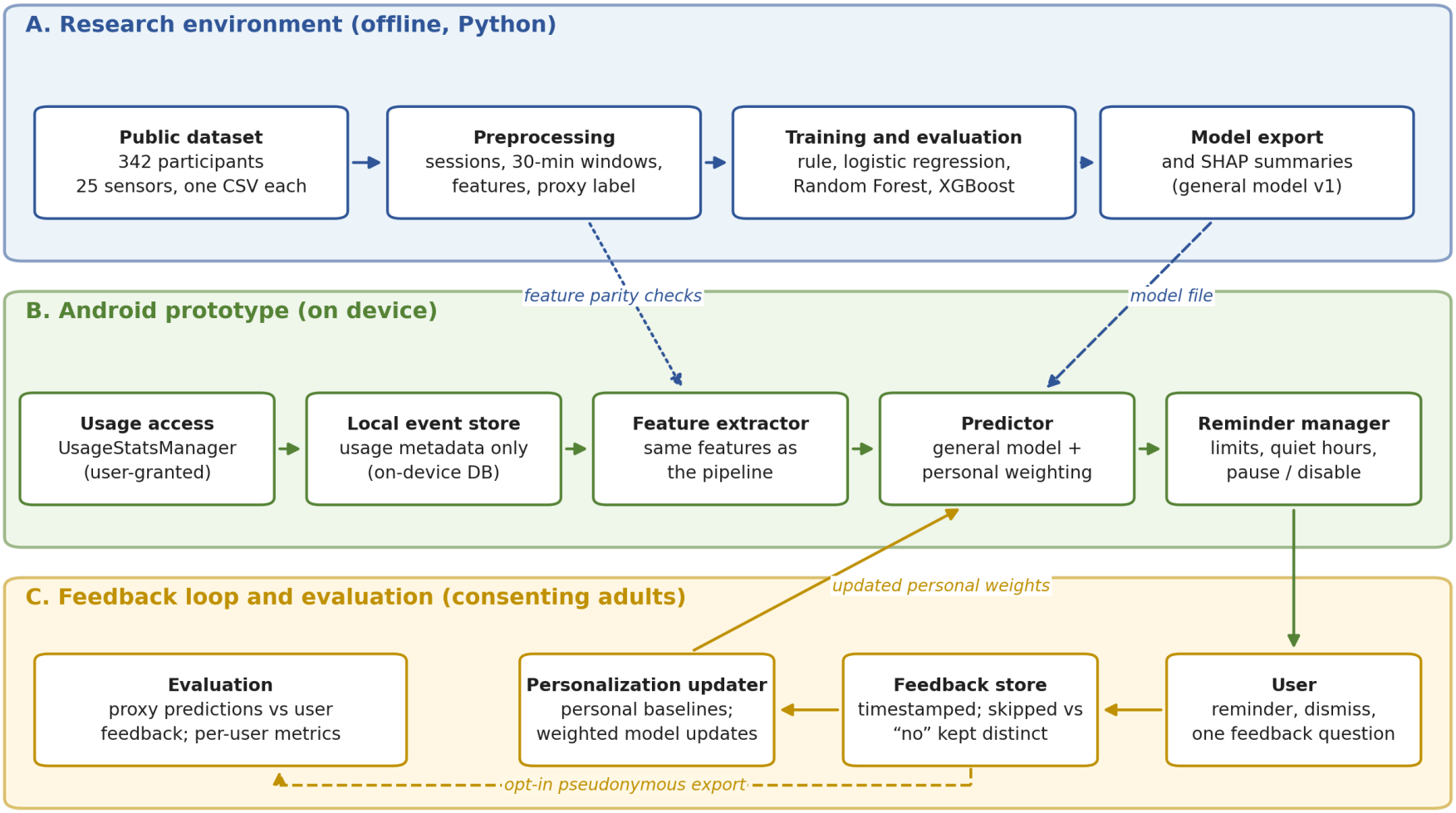


Figure 1. System architecture: (A) offline research pipeline, (B) on-device Android prototype, (C) feedback loop and evaluation.

Table 2. Components of the architecture

Component	Function
A1. Public dataset	Logs of 342 participants; used to build the general model and to test personalization offline.
A2. Preprocessing	Orders events by raw timestamp, reconstructs sessions, flags recording gaps, builds 30-minute windows and features, and assigns the proxy label of the following window.
A3. Training and evaluation	Compares the rule baseline, logistic regression, Random Forest and XGBoost under participant-grouped and chronological splits.
A4. Model export	Stores the selected general model (v1) in a format the app can load, plus SHAP summaries used for plain-language explanations.
B1. Usage access	Reads usage events through UsageStatsManager after the user grants usage access in system settings (Google, n.d.-b).
B2. Local event store	Keeps only usage metadata on the device: timestamps, screen and unlock events, and the foreground-app identifiers needed to count app switches.
B3. Feature extractor	Computes the same features as the Python pipeline for each completed window; parity is checked on shared test cases.
B4. Predictor	Applies the general model to the user's features; personalization enters through personal baselines and, when available, a sample-weighted retrained personalized model or updated decision thresholds.
B5. Reminder manager	Decides whether a reminder is shown; enforces a daily cap and quiet hours and lets the user dismiss, pause or disable reminders.
C1. User	Sees a brief reminder, can dismiss or pause it, and answers one question on whether recent use lasted longer than wanted.
C2. Feedback store	Stores each answer with its time interval and feature record; distinguishes an explicit "no" from a skipped question; stores accepting or dismissing a reminder separately.
C3. Personalization updater	Recomputes personal baselines from local history and applies the sample-weighted retrained model when available. Sample-weighted retraining is evaluated offline first; on-device adaptation will be assessed for latency, memory and battery.
C4. Evaluation	Compares proxy predictions with user feedback and reports per-user metrics and agreement before the two label sources are combined.

6. Data Sources

The development data come from the public Mobile Phone Use Dataset (Pielot, n.d.), released with Pielot et al. (2017). It contains one CSV file for each of 342 participants and records events from 25 sensors, including screen state, foreground application and notifications, together with experience-sampling reports and partial questionnaire data. Our descriptive analysis of the participant table shows that records start between 14 April and 29 June 2016, participants were aged 18 to 66 (M = 37.9) and 329 of 342 used a Spanish locale. The median participant has 21 active days; 280 participants have at least 14 and 220 have at least 21, which supports the planned 14-day history experiments. The dataset contains no unwanted-use label, so we will derive a documented proxy label from long sessions, repeated unlocking and night-time activity. Preprocessing orders events by raw timestamp, reconstructs sessions (Parry & Toth, 2025), flags recording gaps, aggregates 30-minute windows and extracts features that the Android prototype can reproduce. Reuse conditions will be verified before any redistribution. A second, small data source will be usage metadata and brief feedback from consenting adult users of our prototype.

7. Literature Review

Recent work links objective logs and self-report but shows that they capture different things: a meta-analysis found only moderate agreement between logged and self-reported use (Parry et al., 2021), and a methodological primer shows how to reconstruct sessions from Android event logs (Parry & Toth, 2025). Prediction studies support tree-based models on tabular data: Random Forest performed best on questionnaire-derived variables from a large Korean survey (Lee & Kim, 2021), and adding a user's previous responses raised F1 from 0.113 to 0.311 when predicting engagement in 337 users (Pielot et al., 2017). Intervention studies suggest that adaptation and user control matter: Time2Stop adapted reminders to feedback in a field study with 71 users, but effects differed across outcomes and the authors name the cold-start problem as a limitation (Orzikulova et al., 2024), while Socialize let users choose which habits to change (Monge Roffarello & De Russis, 2021). Among the studies we reviewed, none tests how much a model pre-trained on many users helps during the first days of personalization. Our project builds on this work by testing that question with a general, a personalized and a personal-only model, first on proxy labels and later against user feedback.

Implementation Plan

1. Technology Stack

Table 3. Technology stack

Layer	Tools	Role
Programming languages	Python 3; Kotlin	Python for the data pipeline and modelling; Kotlin for the Android prototype.
Data and ML libraries	pandas, NumPy, scikit-learn, XGBoost, SHAP, matplotlib	Preprocessing; logistic regression and Random Forest; XGBoost; grouped cross-validation and metrics (F1, PR-AUC); explanations; figures.
Android platform	Android Studio and SDK; UsageStatsManager; Jetpack Room; WorkManager; notifications; Jetpack Compose	Event collection, local database, scheduled feature extraction, reminders and interface. Screen and keyguard events require Android 9 (API level 28) or later (Google, n.d.-c).
Model deployment	ONNX Runtime Mobile (candidate); exported decision rules (fallback)	On-device inference; the choice is made after a size and latency test.
Development environment	Jupyter and VS Code; Git and GitHub	Shared code and issue tracking; fixed random seeds and versioned data splits for reproducibility.
Hardware	At least two Android phones running different versions (API 28+); laptops	Feasibility and battery tests; CPU is sufficient for tabular models, no GPU needed.
Privacy tooling	In-app consent screen; local-only storage; optional pseudonymous export	Consent and data minimization aligned with KVKK principles.

2. Timeline

The schedule follows the capstone calendar: the concept note (12–18 October), data preparation and feature engineering (19–30 October, with a presentation on 30 October), model development (31 October–10 November), deployment (11–17 November), the draft presentation (18–26 November) and the final presentation on 27 November. Because the deployment week is short, the Android prototype is developed in parallel from the start, so that deployment is integration rather than first-time development. Figure 2 shows the plan and Table 4 lists the tasks.

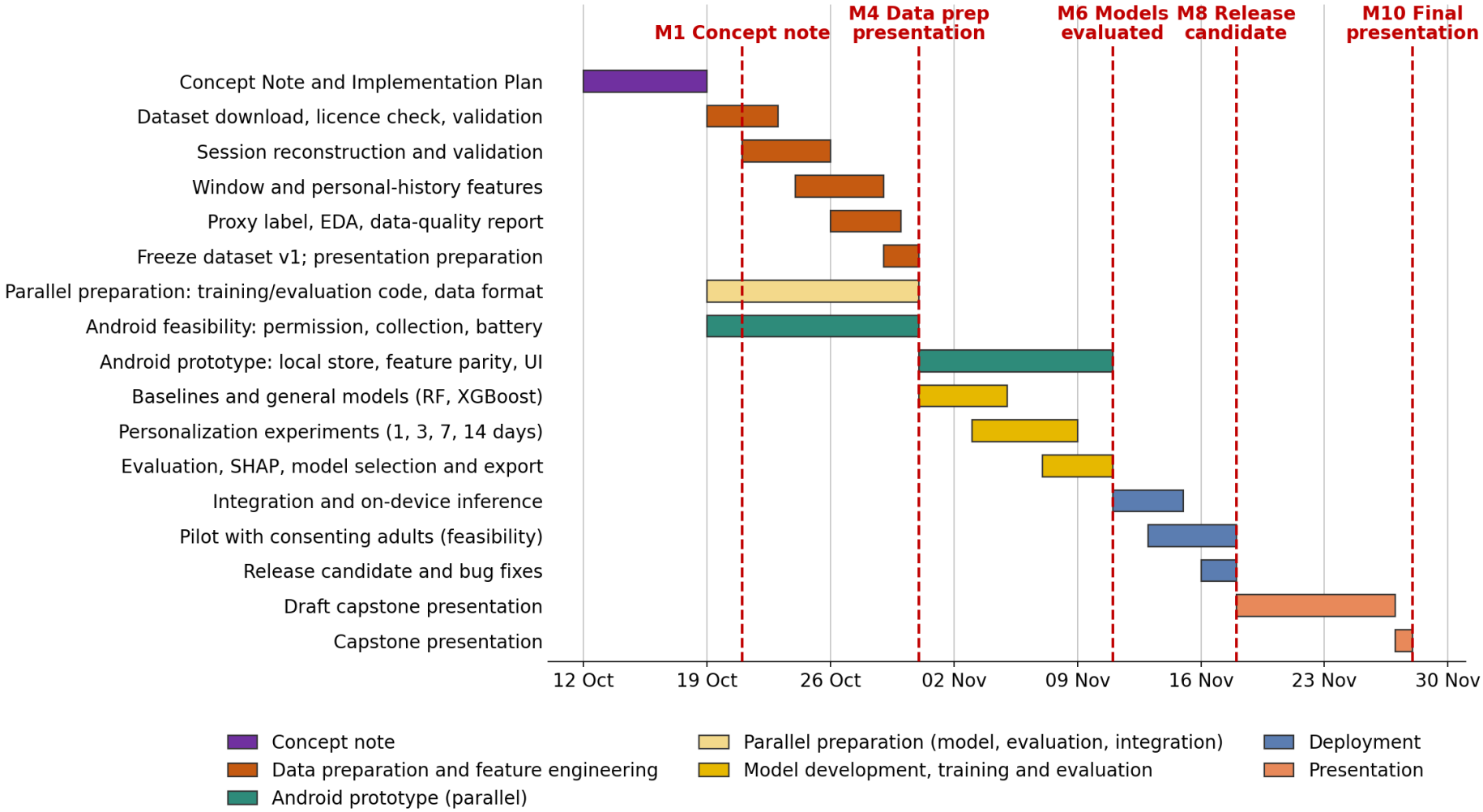


Figure 2. Project timeline (12 October – 27 November 2026) with milestones.

Table 4. Stages and tasks

Stage	Dates	Main tasks
Data collection and preprocessing	19–30 Oct	Complete dataset validation, record licence and reuse terms, validate all 342 files (19–22 Oct); reconstruct and validate sessions (21–25 Oct); build window and personal-history features (24–28 Oct); define proxy labels, run exploratory analysis and write the data-quality report (26–29 Oct); freeze dataset v1 and prepare the presentation (29–30 Oct).
Android feasibility (parallel)	19 Oct–10 Nov	Permission flow, event collection, background reliability and battery test (to 30 Oct); then local store, feature extractor with parity checks and interface (31 Oct–10 Nov).
Model development	31 Oct–4 Nov	Rule and logistic-regression baselines; general Random Forest and XGBoost with participant-grouped cross-validation.
Training and evaluation	3–10 Nov	Personalization experiments (1, 3, 7, 14 days; general, personalized, personal-only); evaluation with per-user results and false alerts per day; SHAP analysis; model selection and export.
Deployment	11–17 Nov	Integrate the model and reminder logic; on-device inference test; pilot with consenting adults; release candidate and bug fixes.
Presentation	18–27 Nov	Draft capstone presentation (18–26 Nov); final presentation (27 Nov).

Parallel start in the first two weeks. So that nobody has to wait for data preparation to finish, all workstreams start together on 19 October:

- **İsmail Hakkı Sarıoğlan (data)** processes a small data sample and prepares the first feature table.
- **Kaan Özsoy (model development)** clarifies the prediction target and the labelling approach with the data lead, sets up the training code and starts first trials on the sample.
- **Hamza Kaya (evaluation)** prepares the participant-grouped and chronological split rules, the metric code and the evaluation checks.
- **Murat Gürel (Android app)** sets up the Android project and works on permissions, data collection and basic screens.
- **Yiğit Kayalı (integration and testing)** defines the common data format and prepares the integration infrastructure with sample inputs and placeholder prediction outputs.

Table 5. Task distribution matrix

Workstream	Dates	Kaan Özsoy	İsmail H. Sarıoğlan	Yiğit Kayalı	Hamza Kaya	Murat Gürel
W1. Data preparation and feature extraction	19 Oct – 30 Oct	S	R	S	S	–
W2. Model development and personalization	19 Oct – 10 Nov	R	S	–	S	–
W3. Model evaluation and experiment design	19 Oct – 10 Nov	S	S	–	R	–
W4. Android app: permissions, data collection, interface	19 Oct – 17 Nov	–	–	S	–	R
W5. Model integration and system tests	19 Oct – 17 Nov	S	–	R	–	S
W6. Ethics, pilot coordination, documentation and presentations	All phases	Shared	Shared	Shared	Shared	Shared

R = responsible (workstream lead); S = supports; Shared = shared by all members. Each member prepares the report and presentation sections of their own workstream; pilot coordination and the merging of the documents are shared.

3. Milestones

Table 6. Milestones and success criteria

ID	Date	Milestone	Success criterion
M1	20 Oct	Concept note and plan submitted	Document submitted with architecture diagram, timeline and task matrix.
M2	22 Oct	Dataset validated	All 342 participant files load; licence and reuse terms recorded, participant inventory (active days, sensor coverage) produced; a common data format is agreed with the integration lead.
M3	27 Oct	Pipeline v1	Reconstructed sessions match hand-checked examples for at least five participants; automated tests confirm that no feature uses data after the window end; the first feature table is handed over to the model and evaluation workstreams.
M4	30 Oct	Data preparation presentation	Dataset v1 frozen; proxy-label prevalence and sensitivity results and the data-quality report presented.
M5	4 Nov	Baselines and general models	Rule, logistic regression, Random Forest and XGBoost compared under participant-grouped cross-validation (PR-AUC, F1, false alerts per day).
M6	10 Nov	Models evaluated	Three-way comparison for 1, 3, 7 and 14 days with per-user results; model selected and exported.
M7	14 Nov	On-device collection stable	Prototype records events for 24 hours on at least two test phones without crashes; battery impact measured and reported.
M8	17 Nov	Release candidate	Integrated prototype with consent screen, capped reminders and feedback question; pilot feasibility reported with the number of consenting participants.

ID	Date	Milestone	Success criterion
M9	26 Nov	Draft presentation	Complete draft rehearsed with the whole team.
M10	27 Nov	Final presentation	Presentation delivered.

4. Challenges and Mitigations

Table 7. Anticipated challenges and mitigations

Area	Challenge	Mitigation
Data quality	Sensors differ between participants and app versions: screen events are On/Off/Unlocked for one participant but only Unlocked/Off for another, and notification actions differ.	Map all variants to a unified event schema; unit tests on hand-checked sessions.
Data quality	Timestamps: the local-time field differs from the raw timestamp and, for experience-sampling rows, reflects posting time.	Order events by raw timestamp and convert to local time explicitly.
Data quality	Recording gaps and implausible sessions, for example a 4.4-hour session and many sessions of about five minutes that may reflect the screen timeout.	Flag gaps; exclude or cap invalid sessions; report sensitivity of results to these choices.
Data quality	Questionnaire data are largely missing (PHQ-8 for 64 of 342 participants).	Do not use questionnaire scores as general features.
Data quality	The dataset is from 2016 and from Spain.	Treat it as development data; check transfer on app data; state the limits.
Model performance	Class imbalance and proxy-label validity: the proxy label covered 3.7–8.8% of windows in a two-participant check.	Use PR-AUC and F1; class weights on training data only; threshold sensitivity; compare with user feedback.
Model performance	Machine learning may not beat simple rules.	Always report the rule baseline; if there is no gain, report the negative result and fall back to personal-percentile thresholds.
Model performance	Optimistic estimates from leakage or random splits.	Participant-grouped and chronological splits; gap between feature and outcome intervals; per-user reporting.
Model performance	Sparse user feedback.	Report label counts alongside elapsed time; present feasibility findings if labels are too few.
Technical	Event availability differs by Android version; screen and keyguard events need API 28 or later (Google, n.d.-c).	Set the minimum supported version accordingly; test on at least two devices.
Technical	Background execution and battery use.	Scheduled jobs with WorkManager; minimal sampling; measure battery impact (M7).
Technical	On-device inference for tree models.	Evaluate ONNX Runtime Mobile; fall back to exported decision rules; decide by 12 Nov.
Schedule	Eleven days for model development and seven for deployment.	Parallel Android work from 19 Oct; feature freeze on 29 Oct; weekly review; small scope.

Area	Challenge	Mitigation
Schedule	The pilot is too short to show 14-day personalization.	Compute personalization curves on the public dataset; use the pilot for feasibility only.

5. Ethical Considerations

Privacy and consent. The pilot involves adults only. Before collection, participants receive an explanation of the data collected, its purpose, storage and how to withdraw, and give informed consent; institutional requirements are addressed before recruitment. Our design follows the principles of lawfulness, purpose limitation, proportionality and limited retention in the Turkish Personal Data Protection Law (Personal Data Protection Authority, n.d.). Collection is limited to usage metadata; screenshots, message contents, contact lists and precise location are out of scope. Participant identifiers are pseudonymous, which we do not describe as full anonymization. Raw logs stay on the device where practical; any research export is opt-in, limited to necessary records, access-restricted and covered by retention and deletion rules set before collection. On-device processing alone is not treated as proof of compliance.

Public dataset. The dataset was released in anonymized form by its authors. We will not attempt to re-identify participants, will verify the reuse conditions before redistributing anything, and will publish only derived features and results.

Bias and representativeness. The dataset comes from Android users in Spain in 2016, aged 18 to 66 with a mean age of about 38, and questionnaire data are partial. It may not represent users in Türkiye, students or today's applications. We therefore report results per user and per age group rather than only averages, and avoid claims for groups the data do not cover.

Impact on users. Reminders can annoy, induce guilt or add interruptions. We limit this with a daily cap, quiet hours and options to dismiss, pause or disable them, and by using neutral, descriptive wording. The system speaks of "elevated use", not addiction; proxy labels are not validated measures of unwanted use, and users decide for themselves which use they did not want (Monge Roffarello & De Russis, 2021).

Professional support and transparency. When a user reports difficulty controlling their use, the app shows information about professional support, such as the counselling services of Yeşilay's YEDAM (Yeşilay, n.d.), and states that it is not a diagnostic or therapeutic tool. Explanations describe observed behaviour and do not claim causes. We will report limitations and negative results openly.

6. References

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